

Monitoring

Monitoring in an Azure Landing Zone is the continuous collection, analysis, and alerting of the entire system to ensure Azure resources are available, healthy, secure, performing, cost-effective, and compliant

When we monitor an end-to-end system, we must consider infrastructure monitoring for components such as VMs, VMSS, AKS, SQL databases, storage accounts, App Services, Function Apps, Logic Apps, and Web Apps.

We will also consider network monitoring, which includes VNet/Subnet flow logs, ExpressRoute, VPN, Load Balancers, Application Gateways, and Front Door

Identity Monitoring: Additionally, we will include identity monitoring, which encompasses Microsoft Entra-ID logs, MFA logs, PIM logs, RBAC logs, and Conditional Access logs.

Furthermore, we will implement security monitoring, focusing on the integration of Microsoft Defender for Cloud across various Azure services

we will include backup and disaster recovery monitoring to track whether Azure Backup and Azure Site Recovery tasks succeeded or failed

We will also incorporate cost optimization tracking and application-level logging

For our monitoring solution, we have two options: an Azure-native solution and an open-source solution

Azure-Native Solution

Open-Source Solution

The aim of Azure monitoring is to create a centralized logging system that collects metrics, logs, events, and alerts across the entire environment

For collecting and storing logs, we utilize a Log Analytics Workspace, which organizes and stores data in a structured table and column format

For example, it stores logs from AKS, VMs, Azure Firewall, VMSS, and other resources.

Once the logs are collected, everything is queried using KQL (Kusto Query Language).

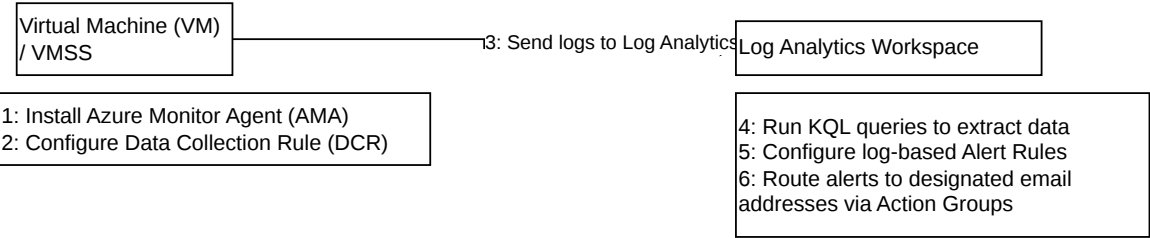
For applications, we utilize Application Insights, which tracks failures, response times, performance metrics, request rates, and active user sessions

Every monitoring tool serves two core functions:

- 1. **Logs:** Tracking Activity logs, Subscription logs, and Diagnostic logs.
- 2. **Metrics:** Measuring CPU, Memory, Disk usage, Latency, and Availability.

Most monitoring tools operate in a similar fashion: a monitoring agent is installed on the resources that need to be monitored.

In Azure Monitor, this agent is called the **Azure Monitor Agent (AMA)**. The AMA is responsible for collecting Windows event logs, Linux syslogs, performance counters, and custom logs. It relies on **Data Collection Rules (DCRs)** to define exactly what data to collect, such as specifying whether to gather syslogs or event logs.



Infrastructure monitoring includes tracking CPU utilization, memory usage, disk I/O, resource availability, agent heartbeats, running processes, OS events, and core system services

Infrastructure monitoring tracks CPU utilization, memory usage, disk I/O, resource availability, agent heartbeats, running processes, OS events, and system services
For example, an alert rule can be configured to trigger when CPU utilization exceeds 70%.

// Alert rule when CPU utilization is greater than 70% over the last 5 minutes

```
InsightsMetrics
| where Origin == "vm.azm.ms"
| where Namespace == "Processor" and Name == "UtilizationPercentage"
| summarize AvgCPU = avg(Val) by ResourceId, bin(TimeGenerated, 5m)
| where AvgCPU > 70
```

// Alert when a user has more than 3 failed Entra ID logins within 15 minutes

```
SigninLogs
| where TimeGenerated > ago(15m)
| where ResultType != 0 // 0 means success, any other code means failure
| summarize FailedCount = count() by UserPrincipalName, IPAddress
| where FailedCount > 3
```

Best Practices for azure monitoring

With the KQL query reports, we can pin the visualizations to an Azure Dashboard, Azure Managed Grafana, or Power BI, ensuring the landing zone dashboard includes comprehensive resource health insights.

As a best practice, always develop a centralized logging system within a dedicated Log Analytics Workspace.

As a best practice, always configure Data Collection Rules (DCRs) to pull only the most critical logs instead of pushing all logs to the workspace.

Always enable diagnostic settings for all critical resources.

Always create an Action Group to integrate alerts with communication platforms such as Microsoft Teams, Slack, email, or any other third-party incident management tools

Create a dashboard for critical resources to monitor both health and cost metrics.

To prevent budget overruns, we will establish an Azure Budget policy with multi-tiered alert thresholds based on both actual expenditures and forecasted costs, routing immediate notifications to the platform team via Action Groups.

Always monitor backup status, disaster recovery readiness, compliance, governance, costs, and resource health continuously across all resources.

